PROGRAM : CompTia A+

ASSESSMENT : CODE AP-A1

Hardware Components & System Assembly

A Section : A — Concept Application

S 1

SCENARIO: You are setting up a new office workstation and need to choose the core hardware for it from a supplier's parts list.

Question: Explain the process of selecting a compatible motherboard, CPU, RAM, and storage device for a new office workstation, and justify how you would confirm they will work together.

ANS: 1. Motherboard: Select a laptop motherboard that supports the required processor, RAM, and storage.

2. CPU: Choose an Intel Core i5 or AMD Ryzen 5 processor for office work.

3. RAM: Select 8GB or 16GB RAM for smooth multitasking.

4. Storage: Choose a 512GB SSD for faster booting and file access.

5. Compatibility Check: Check the laptop manufacturer's specifications to confirm CPU, RAM, and SSD compatibility. Verify that all components are detected properly.

For office use, a laptop with an Intel Core i5, 16GB RAM, and 512GB SSD is a good choice for daily office tasks, MS Office, browsing, and multitasking.

S 2

SCENARIO :You are advising a client whose employees mainly run office and productivity applications on their workstations.

Question: Explain the differences between HDD, SSD, and NVMe storage, and recommend the most suitable option for this workload, giving technical reasons for your choice.

ANS : 1. HDD (Hard Disk Drive): Uses magnetic disks to store data. It is cheaper but slower.

2. SSD (Solid State Drive): Uses flash memory. It is faster, quieter, and more reliable than HDD.

3. NVMe SSD: Uses PCIe technology and provides higher speed than a normal SSD.

Recommendation:

I my opinion an NVMe SSD for office workstations because it provides fast booting, quick application loading, and smooth multitasking.

Conclusion: NVMe SSD is the best option for modern office work due to its speed, performance, and reliability.

S 3

SCENARIO: You are configuring a workstation that will run mildly graphics-intensive design software for several hours a day.

Question: Explain the role of the GPU and the cooling system in maintaining performance and stability under this kind of sustained load.

ANS: 1. GPU (Graphics Processing Unit):

- Handles graphics and visual processing.
- Improves the performance of design software.
- Provides smooth images and faster rendering.

2. Cooling System:

- Keeps the CPU and GPU at safe temperatures.
- Prevents overheating during long working hours.
- Maintains system performance and stability.

S 4

SCENARIO: You are assembling a PC from components sourced from three different vendors.

Question: Explain why hardware compatibility matters when assembling a computer, and identify potential issues that could arise from incompatible components.

ANS: Hardware compatibility means ensuring that all computer components work properly together.

Importance of Compatibility:

- Ensures smooth computer performance.
- Prevents hardware damage.
- Improves system stability and reliability.

Possible Issues:

1. CPU may not fit the motherboard socket.
2. RAM may not be supported by the motherboard.
3. Storage devices may not be compatible.
4. PC may fail to start or may crash frequently.
5. Components may overheat or perform poorly.

S 5

SCENARIO: You are finalizing the build and need to connect the power supply unit (PSU) to the motherboard, GPU, and storage devices before first boot.

Question: Explain how you would choose an appropriately rated PSU for this build and describe the safety and connection checks you would carry out before powering it on.

ANS: 1. Choosing the PSU:

- Choose a power supply with enough wattage.
- Select a good-quality PSU.
- Make sure it has the correct cables for all components.

2. Safety Checks:

1. Switch off and unplug the power cable.
2. Connect the motherboard power cable.
3. Connect the CPU and GPU power cables.
4. Connect the storage power cables.
5. Check all wires properly.
6. Make sure everything is connected before turning on the PC.

S 6

SCENARIO :You are handing over a freshly assembled workstation to the client and need to confirm it is ready for use.

Question: Explain the checks you would perform to confirm the system passes POST successfully, and what you would do if it did not.

ANS: 1. POST Checks:

- Turn on the computer.
- Check if the power light is on.
- Check if the monitor displays anything.
- Check if RAM, CPU, and storage are working.
- Make sure there are no error sounds or messages.

2. If POST Fails:

1. Check all cables.
2. Remove and fit RAM again.
3. Check CPU and GPU connections.
4. Check the monitor cable.
5. Restart the computer.

B Section B — Practical Coding Tasks

Task 1: Component Identification & Functions Identify and document the functions of the core hardware components in a desktop system. (Foundational — direct identification of a single concept.)

- **Identify and describe the function of the motherboard, CPU, RAM, GPU, storage device, and cooling system.**
- **Note the make/model or key specification of each component where visible.**
- **Present the findings as a labeled component list or table.**

ANS: The following are the main hardware components of a desktop computer and their functions.

Component	Function	Example / Specification
Motherboard	Connects all computer components.	ASUS / MSI
CPU	Processes instructions and performs calculations.	Intel Core i5
RAM	Temporarily stores data while programs are running.	8GB / 16GB DDR4
GPU	Displays images, videos, and graphics.	NVIDIA GeForce

Storage (SSD/HDD)	Stores files, software, and the operating system.	512GB SSD / 1TB HDD
Cooling System	Keeps components cool and prevents overheating.	CPU Fan / Cabinet Fan

Task 2: Motherboard Ports & Socket Identification Identify the motherboard's physical layout, ports, and CPU socket type. (Applied — one concept examined in detail across multiple points.)

- **Identify and label all major motherboard ports (SATA, power, USB headers, etc.).**
- **Identify all expansion slots (PCIe, M.2, etc.) and their purpose.**
- **Identify the CPU socket type and note which CPU families are compatible with it.**
- **Photograph or diagram the motherboard with labels.**

ANS: 1. Major Motherboard Ports

Port	Purpose
SATA	Connects HDD and SSD.
USB Header	Connects front USB ports.
24-pin Power	Supplies power to motherboard.
CPU Power	Supplies power to CPU.
HDMI	Connects monitor.
Ethernet (LAN)	Connects network cable.
Audio Port	Connects speakers and headphones.

2. Expansion Slots

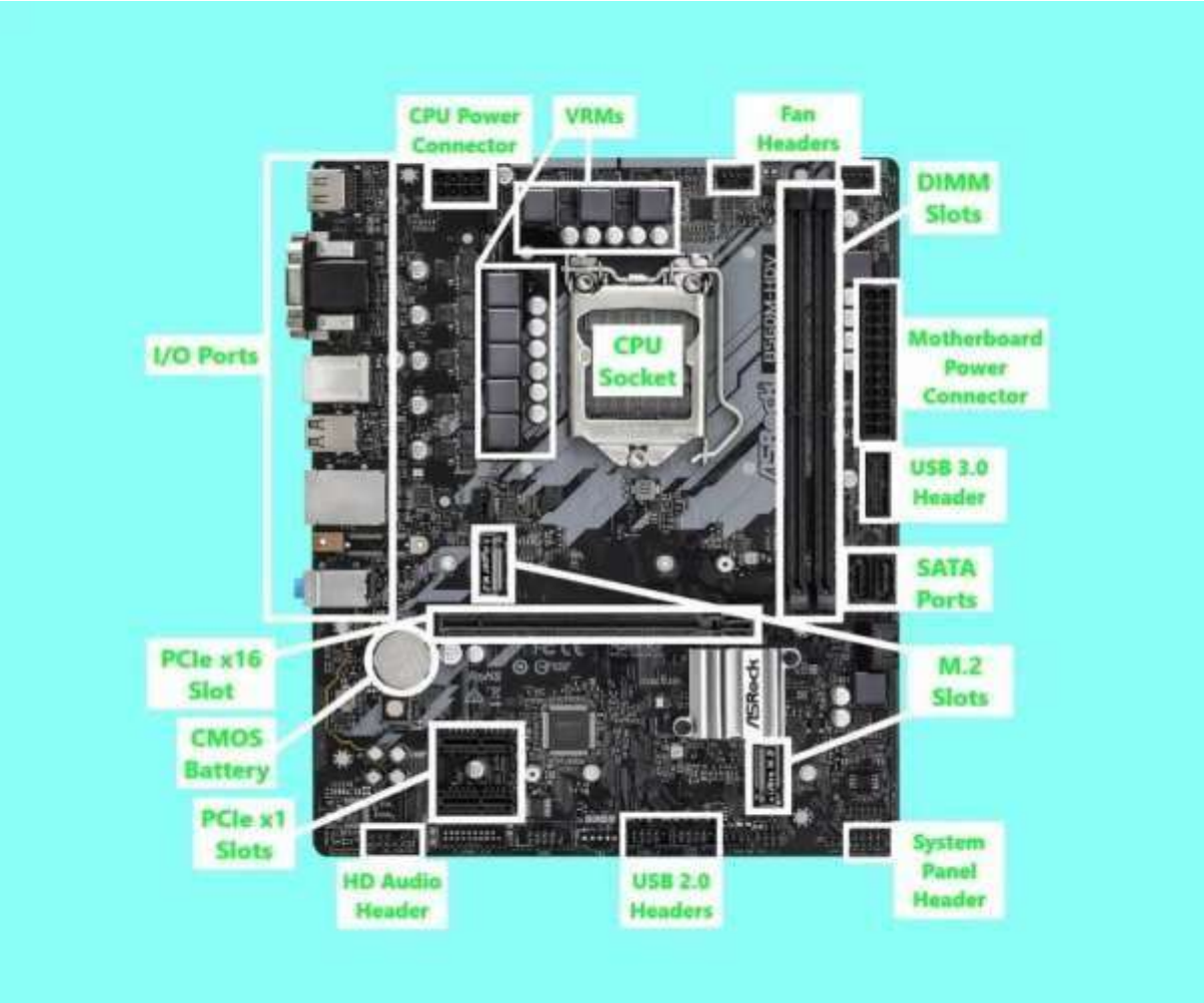
Slot	Purpose
PCIe x16	Connects graphics card (GPU).
PCIe x1	Connects Wi-Fi and sound cards.

M.2 Slot Connects NVMe SSD.

RAM Slots Hold memory modules.

3. CPU Socket Type

- The CPU socket holds the processor.
- Intel processors use sockets such as LGA1700.
- AMD Ryzen processors use sockets such as AM4 and AM5.
- The CPU must match the motherboard socket and supported CPU list. 4. Motherboard Diagram



Task 3: RAM & Storage Installation Remove and reinstall a RAM module and a storage device. (Intermediate — two components handled together with correct procedure.)

- **Safely remove the RAM module, noting ESD precautions taken.**

- **Reinstall the RAM module correctly and confirm it is seated properly. Remove and reinstall the storage device, noting the connection type used (SATA/NVMe).**
- **Confirm both components are detected correctly after reinstallation.**

ANS: 1. RAM Installation

1. Switch off the computer and unplug the power cable.
2. Wear an anti-static wrist strap to prevent ESD damage.
3. Open the RAM slot clips and remove the RAM carefully.
4. Match the RAM notch with the slot.
5. Push the RAM gently until it clicks into place.
6. Check that the RAM is properly fitted.

2. Storage Installation

1. Switch off and unplug the computer.
2. Remove the storage device carefully.
3. Check the connection type (SATA or NVMe).
4. Reinstall the storage device properly.
5. Tighten the screws if required.
6. Connect the cables if using SATA storage.

3. Checking Components

- Start the computer.
- Open BIOS settings.
- Check whether RAM capacity is displayed.
- Check whether the SSD or HDD is detected.

Task 4: Full Desktop PC Assembly Assemble a desktop PC from the provided components and document the process. (Integrative — combines all Section B skills into one build.)

- **Assemble the PC using the provided motherboard, CPU, RAM, storage, GPU, and PSU.**
- **Follow correct installation sequence and ESD/safety precautions throughout.**

- **Document each step with labeled photos or diagrams.**
- **Power on the system and confirm it passes POST successfully.**

ANS: 1. Safety Precautions

1. Switch off and unplug the power supply.
2. Use an anti-static wrist strap.
3. Handle components carefully.
4. Keep the working area clean.

2. PC Assembly Steps

1. Install the CPU on the motherboard.
2. Apply thermal paste and install the CPU cooler.
3. Insert RAM into the correct slots.
4. Install the SSD or HDD.
5. Fix the motherboard inside the cabinet.
6. Install the GPU in the PCIe slot.
7. Connect the PSU power cables.
8. Connect the storage and other required cables.
9. Close the cabinet and connect the monitor, keyboard, and mouse.

3. PC Assembly Diagram



4. Power On and POST Check

- Connect the power cable and switch on the PC.
- Check whether the fans are running.
- Check whether the display appears on the monitor.
- Open BIOS and confirm CPU, RAM, and storage are detected.
- Make sure there are no error messages.

C Section C — Mini Capstone Project

Mini Project: Workstation Deployment & Boot-Failure Diagnosis Objective:

ABC Solutions is setting up a new computer lab. After assembling several systems, some computers fail to start while others display hardware detection errors, and the IT team suspects incorrect installation or compatibility issues. Build (or inspect) a workstation, diagnose the likely causes of these failures, and produce a diagnostic and installationverification report — combining component selection, installation procedure, and compatibility troubleshooting from Section B.

Your project must:

- **Identify the three hardware components you would check first when a newly assembled PC fails to boot, and explain why.**
- **Explain how motherboard–CPU compatibility affects system functionality, referencing the specific system you assembled or inspected.**
- **Describe how you would verify that RAM and storage devices are installed correctly, including what BIOS/POST should confirm.**
- **Produce a hardware-installation checklist of at least three steps to prevent similar failures in future deployments.**
- **Provide a technical justification for using SSDs instead of HDDs in modern office workstations, tying it back to the lab's use case.**

ANS:

Mini Project: Workstation Deployment & Boot-Failure Diagnosis

Company Name: ABC Solutions

1. Three Hardware Components to Check First

1. RAM: Check whether RAM is properly inserted.
2. PSU: Check whether the power supply is working.
3. CPU: Check whether the processor is installed correctly.

2. Motherboard and CPU Compatibility

The CPU must match the motherboard socket. For example, Intel Core i5-12400 requires an LGA1700 motherboard with BIOS support. If they are incompatible, the computer may not start.

3. RAM and Storage Checking • Check RAM is fitted properly.

- Check SSD or HDD cables.
- Start the computer and open BIOS.
- Check whether RAM and storage are detected.
- Check for any error messages.

4. Hardware Installation Checklist

1. Switch off the computer before installation.
2. Check CPU and motherboard compatibility.
3. Install RAM and storage correctly.
4. Connect all power cables.
5. Check all connections before starting.
6. Start the computer and check POST.

5. Why SSD Is Better Than HDD

- SSD is faster than HDD.
- SSD starts the computer quickly.
- SSD opens applications faster.
- SSD has no moving parts.
- SSD is suitable for office work.

Recommendation: Use a 512GB SSD for office computers because it provides fast performance and saves time.

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D Section D — AI-Augmented Learning

STEP 1 · BUILD WITH AI Use an AI tool of your choice (ChatGPT, Claude, Copilot, etc.) to help you produce a step-by-step troubleshooting checklist that:

- **Guides a junior technician through diagnosing a desktop PC that fails to boot immediately after assembly, in the correct priority order.**
- **Covers at minimum the motherboard, RAM, storage, and PSU as possible failure points.**
- **Is presented as a clear checklist or simple decision flow a technician could follow on-site.**
- **Is applied to the ABC Solutions scenario from Section C to show which step would have caught the reported issue.**

STEP 2 · TEST & DEBUG (WITHOUT AI) Then, working without AI, review the checklist and find at least one step that is missing, incorrectly ordered, or unsafe (for example, skipping ESD precautions or checking software before physical connections). Fix it yourself

ANS: STEP 1 · BUILD WITH AI

AI Tool Used: ChatGPT

1. Exact Prompt Given to AI

"Create a simple step-by-step checklist to find why a desktop computer is not starting after assembly. Include motherboard, RAM, storage, and PSU checks. Also explain how to fix the problems."

2. AI's Original Checklist

1. Check the power cable and PSU switch.
2. Check motherboard power connections.
3. Check whether RAM is fitted properly.
4. Check SSD or HDD connections.
5. Check CPU installation.
6. Start the PC and check POST and BIOS.
7. Check whether all components are detected.

ABC Solutions: This checklist helps find loose cables, RAM problems, and storage connection problems.

3. My Corrected Checklist (Without AI) 1. Switch off and unplug the computer.

2. Use an anti-static wrist strap.
3. Check the power cable and PSU switch.
4. Check motherboard power cables.
5. Check RAM is fitted properly.
6. Check SSD or HDD connections.
7. Check CPU compatibility.
8. Start the computer and check POST.
9. Open BIOS and check RAM and storage.

4. What I Changed and Why

- I added safety steps before touching computer parts.
- I added unplugging the power cable.
- I added an anti-static wrist strap to protect hardware.
- The AI checklist missed these safety steps, so I corrected them.